20 MAGNETISM AND ELECTROMAGNETISM

There are two types of magnets that we use in our everyday lives. These are permanent magnets and electromagnets. A permanent magnet has a magnetic field around it all the time. The strength and direction of this field is not easy to change. An electromagnet when turned on also has a magnetic field around it but its strength and direction can be changed very easily. In this chapter you will learn about the factors affecting the magnetic field around an electromagnet and how electromagnets are used in several important devices.



▲ Figure 20.1 Electromagnets can be used to lift iron or steel objects.

The huge electromagnet in Figure 20.1 is being used in a scrapyard to pick up large objects that contain iron or steel. When the objects have been moved to their new position the electromagnet is turned off and the objects fall.

LEARNING OBJECTIVES

- Know that magnets repel and attract other magnets and attract magnetic substances
- Describe the properties of magnetically hard and soft materials
- Practical: investigate the magnetic field pattern for a permanent bar magnet and between two bar magnets
- Understand the term magnetic field line
- Know that magnetism is induced in some materials when they are placed in a magnetic field
- Describe how to use two permanent magnets to produce a uniform magnetic field pattern
- Know that an electric current in a conductor produces a magnetic field around it

PHYSICS ONLY

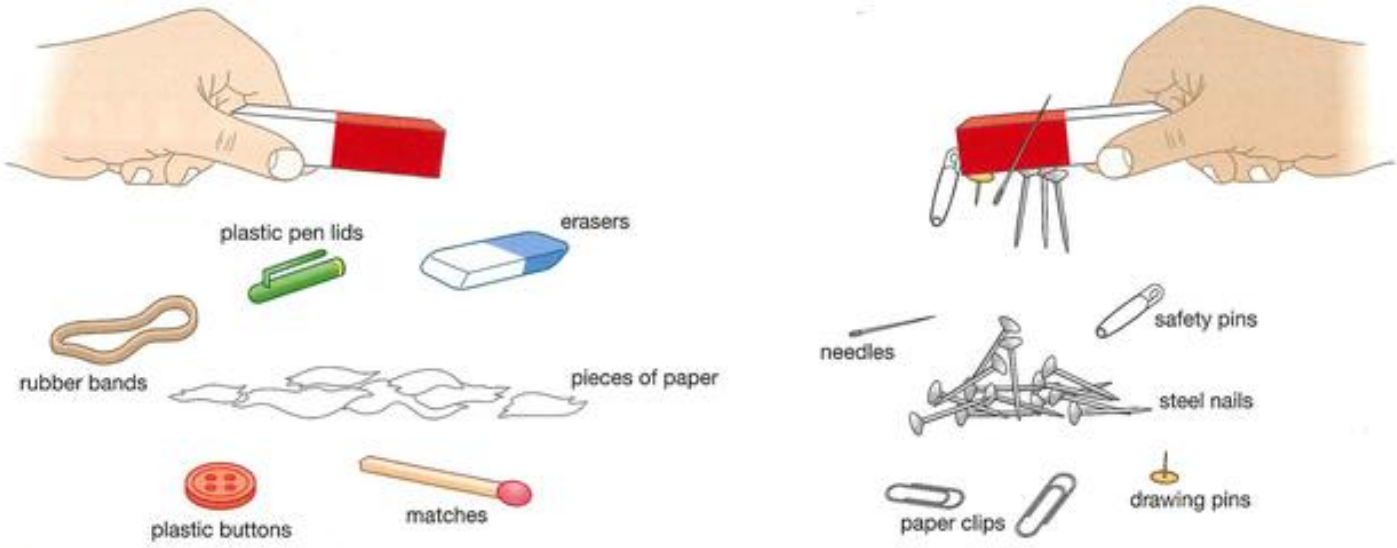
- Describe the construction of electromagnets
- Draw magnetic field patterns for a straight wire, a flat circular coil and a solenoid when each is carrying a current

UNITS

In this section you will need to use ampere (A) as the unit of current, volt (V) as the unit of voltage and watt (W) as the unit of power.

MAGNETISM AND MAGNETIC MATERIALS

Magnets are able to attract objects made from magnetic materials such as iron, steel, nickel and cobalt. Magnets cannot attract objects made from materials such as plastic, wood, paper or rubber. These are non-magnetic materials.

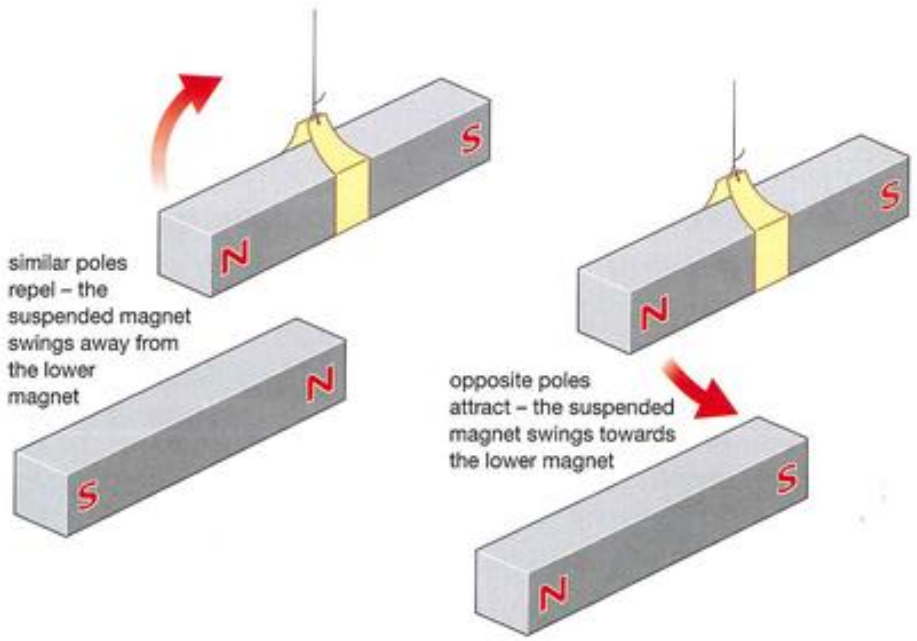


▲ Figure 20.2 Magnets attract some objects and not others.

MAGNETS

The strongest parts of a magnet are called its poles. Most magnets have two poles. These are called the north pole and the south pole.

If two similar poles are placed near to each other they repel. If two dissimilar (opposite) poles are placed near to each other they attract.



▲ Figure 20.3 Similar poles repel and opposite poles attract.

Permanent magnets like the bar magnets shown in Figure 20.3 are made from a magnetically hard material such as steel. A magnetically hard material keeps its magnetism once it has been magnetised. Iron is a magnetically soft material and is not suitable for a permanent magnet. Magnetically soft materials lose their magnetism easily and are therefore useful as temporary magnets.

EXTENSION WORK

Ancient travellers would construct a basic compass by placing a splinter (thin piece) of magnetised rock called lodestone onto a piece of wood floating in water. The splinter would turn to point in the direction of north–south. The end pointing towards the north was called the north pole.

MAGNETIC FIELDS

Around every magnet there is a volume of space where we can detect magnetism. This volume of space is called a **magnetic field**. Normally a magnetic field cannot be seen but we can use iron filings or plotting compasses to show its shape and discover something about its strength and direction.

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE THE MAGNETIC FIELD PATTERNS OF BAR MAGNETS

1 Place a bar magnet between two books and place a sheet of paper or thin card over it.

2 Sprinkle some iron filings on the paper above the magnet.

3 Tap the paper very gently.

4 The iron filings will move to show the magnetic field pattern.

OR

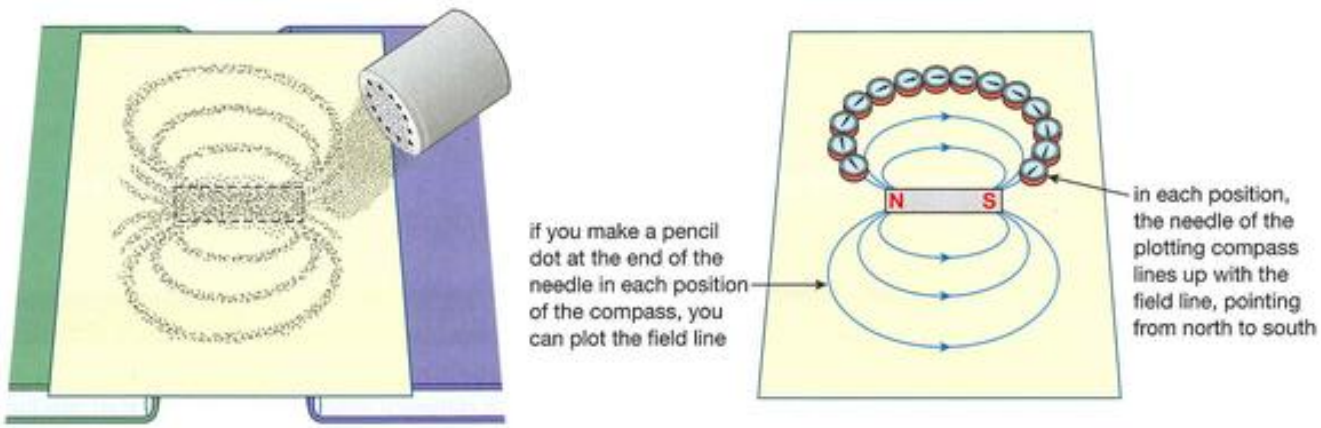
1 Place a bar magnet on a piece of paper.

2 Place a large number of small compasses on the paper near the magnet.

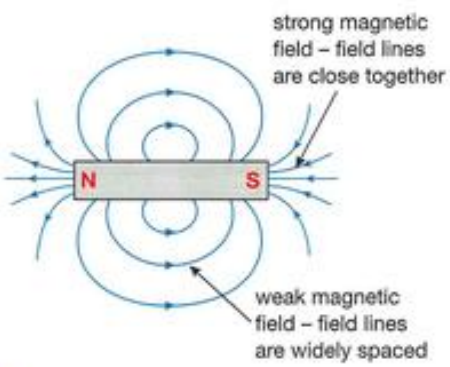
3 Look carefully at the pattern shown by the needles of the compasses.

Repeat the same experiment using two bar magnets, placing them about 5 cm apart.

! Avoid skin contact with iron filings, since they irritate the skin. Do not blow them off the paper, because they may get into the eyes, which is very painful.



▲ Figure 20.4 We can see the shape of the magnetic field around a magnet by using iron filings or plotting compasses.



▲ Figure 20.5 The magnetic field around a bar magnet follows a pattern like this.

We draw magnetic fields like that in Figure 20.5 using magnetic field lines. Magnetic field lines don't really exist but they help us to visualise the main features of a magnetic field.

The magnetic field lines:

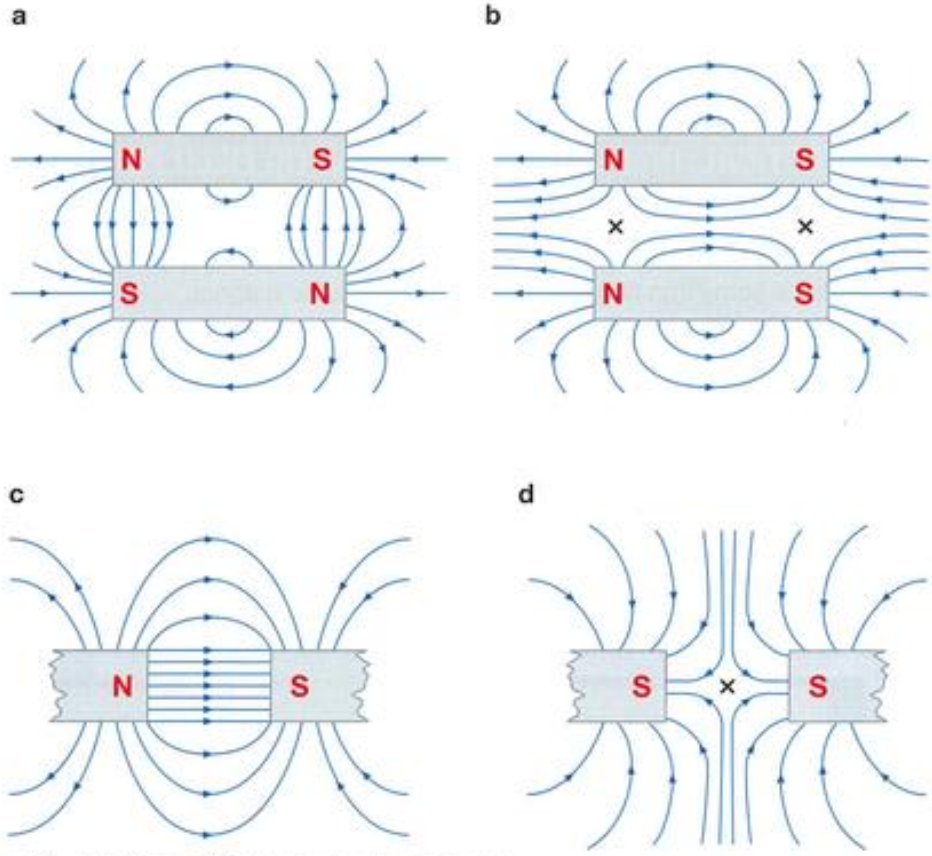
- show the shape of the magnetic field
- show the direction of the magnetic force – the field lines 'travel' from north to south
- show the strength of the magnetic field – the field lines are closest together where the magnetic field is strongest.

OVERLAPPING MAGNETIC FIELDS

If two magnets are placed near each other, their magnetic fields overlap and affect each other. We can investigate this using iron filings or plotting compasses. Figure 20.6 shows the different field patterns we would see.

EXTENSION WORK

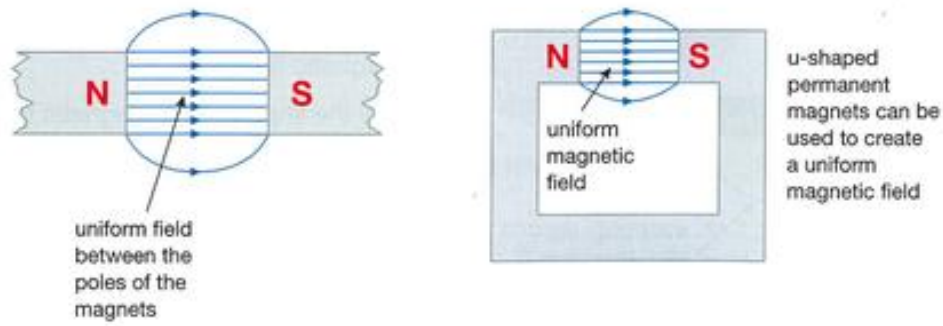
Point X in Figure 20.6 is called a neutral point. It is a position where the effects of the two magnetic fields have cancelled each other out, meaning that there is no resultant magnetic field here. Can you find any other neutral points in these magnetic field patterns?



▲ Figure 20.6 Magnetic fields around pairs of magnets

CREATING A UNIFORM MAGNETIC FIELD

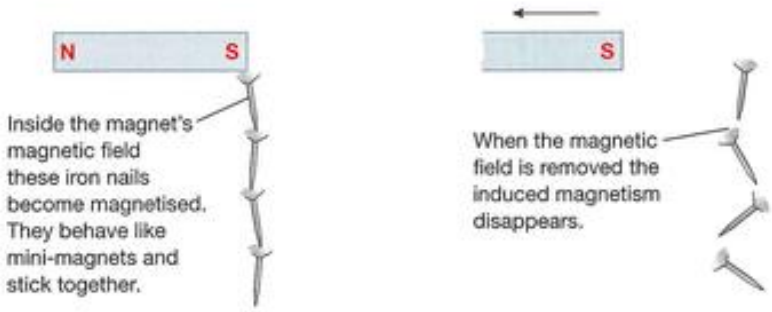
If we look very carefully at the magnetic field created between the north pole of one magnet and the south pole of a different magnet we will see that the field is shown as a series of straight field lines that are evenly spaced. A field like this is described as a uniform magnetic field – that is, its strength and direction is the same everywhere. Creating a uniform field like this is extremely useful as we will see later in this chapter and in the next chapter.



▲ Figure 20.7 Two examples of uniform magnetic fields

INDUCED MAGNETISM

If we place an object made from a magnetic material, for example, an iron nail, inside a magnetic field it becomes a magnet. We say that magnetism has been induced in the nail. Because iron is a magnetically soft material, its induced magnetism is temporary and disappears if the permanent magnet is removed. If the nail is made from a magnetically hard material such as steel it will retain some of its magnetism after the magnet is removed.

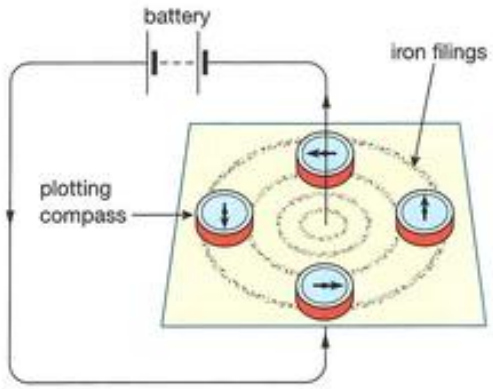


▲ Figure 20.8 Introducing magnetism in iron nails

PHYSICS ONLY

ELECTROMAGNETISM

When there is a current in a wire a magnetic field is created around it. This is called electromagnetism. The field around the wire is quite weak and circular in shape.

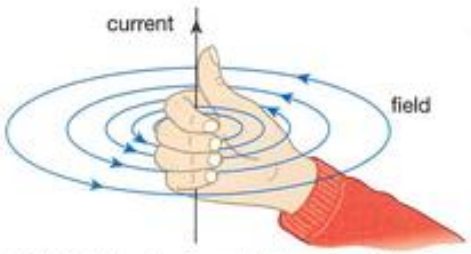


The shape and direction of the magnetic field around a current-carrying wire can be seen using iron filings and plotting compasses. Changing the direction of the current, changes the direction of the magnetic field.

▲ Figure 20.9 A current-carrying wire has a magnetic field around it.

EXTENSION WORK

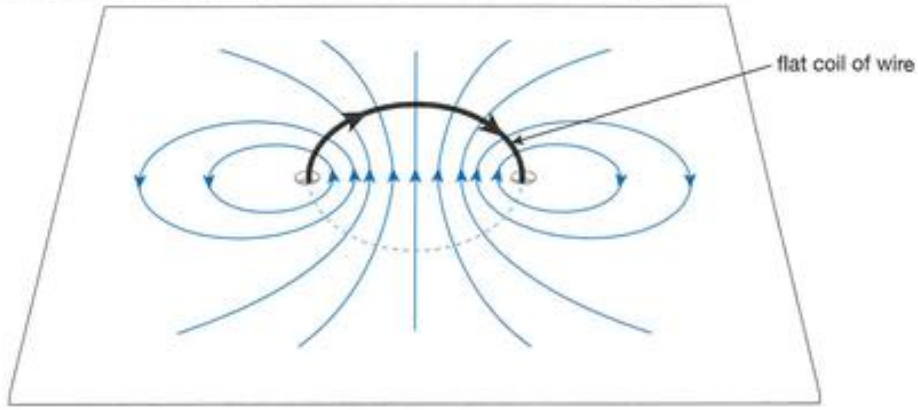
The direction of the magnetic field depends upon the direction of the current. It can be found using the right-hand grip rule (for fields).



With the thumb of your right hand pointing in the direction of the current, your fingers will curl in the direction of the field.

▲ Figure 20.10 You can work out the direction of the field using the right-hand grip rule (for fields).

If the wire is made into a flat, single-turn coil (circular wire), the magnetic field around the wire changes shape and is as shown in Figure 20.11.

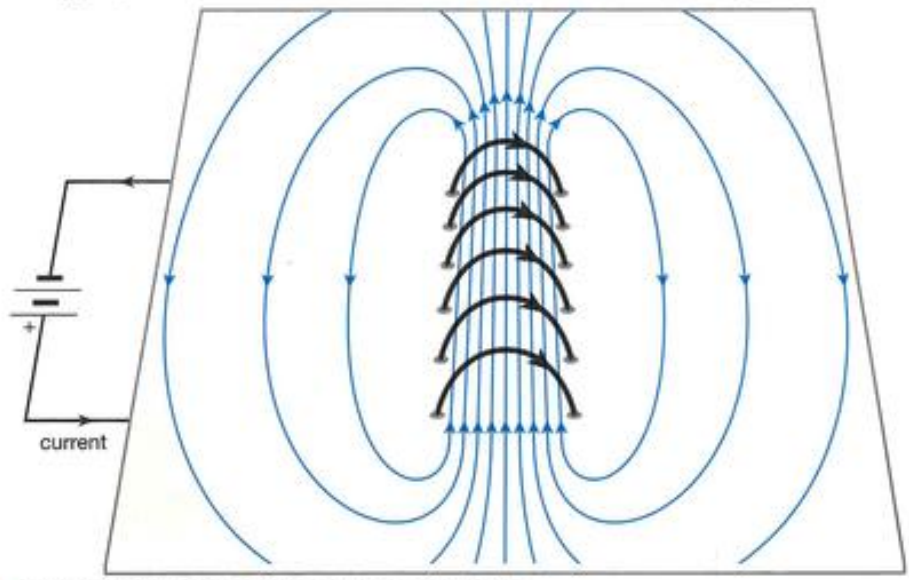


▲ Figure 20.11 The magnetic field around a flat coil

The strength of the magnetic field around a current-carrying wire can be increased by:

1 increasing the current in the wire

2 wrapping the wire into a coil or **solenoid** (a solenoid is a long coil).



▲ Figure 20.12 The field around a solenoid looks like this.

EXTENSION WORK

You will not be asked questions about this but it is included in more advanced Physics courses.
The positions of the poles for a solenoid can be found using the right-hand grip rule (for poles).
If the direction of the current in the solenoid is reversed, so too are the positions of the poles.

The shape of the magnetic field around a solenoid is the same as that around a bar magnet.



▲ Figure 20.13 You can work out the **polarity** of the solenoid by imagining that your right hand is wrapped around it. Your fingers point in the direction of the current and your thumb points to the north pole of the solenoid. This is the right-hand grip rule (for poles).

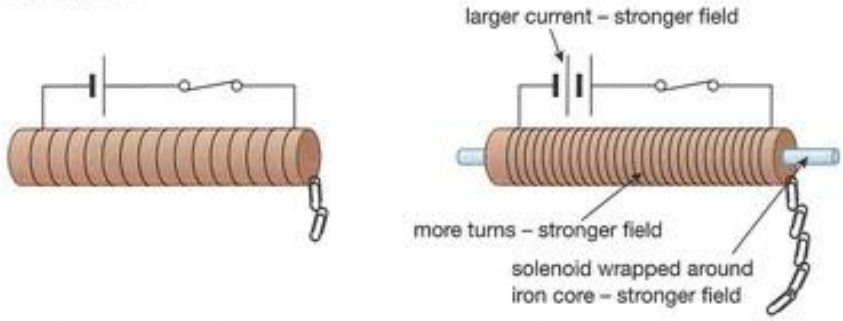
The strength of the field around a solenoid can be increased by:

1 increasing the current in the solenoid

2 increasing the number of turns on the solenoid

3 wrapping the solenoid around a magnetically soft core such as iron.

This combination of soft iron core and solenoid is often referred to as an electromagnet.



▲ Figure 20.14 Altering the strength of the field around an electromagnet

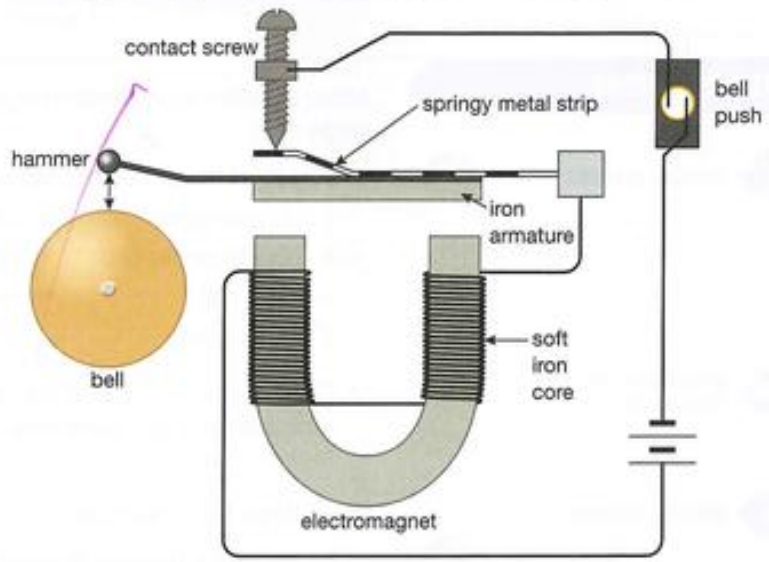
END OF PHYSICS ONLY

EXTENSION WORK

It is not necessary to memorise how an electric bell or a circuit breaker works but understanding how they work might help reinforce some of the basic principles of magnets and electromagnets you will need in your exams.

The electric bell

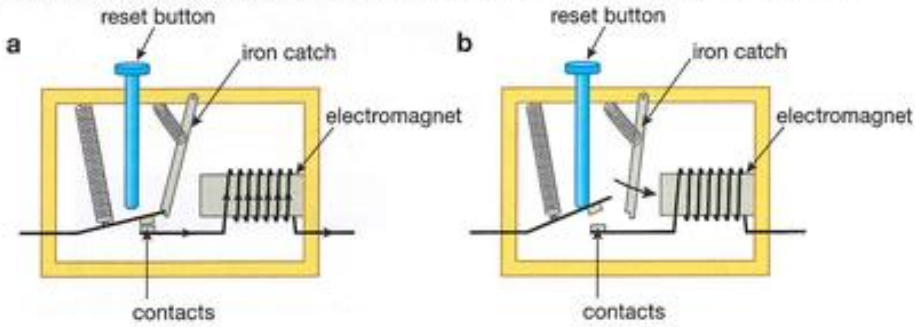
When the bell push is pressed the circuit is complete and there is current in the circuit. As a result the soft iron core of the electromagnet becomes magnetised and attracts the piece of iron that is attached to the hammer (this is called the armature). When the armature is pulled down, the hammer strikes the bell. At the same time a gap is created at the contact screw. The circuit is now incomplete and the current stops. The electromagnet is now turned off so the springy metal strip pulls the armature up to its original position.
The circuit is again complete and the whole process begins again.



▲ Figure 20.15 An electric bell relies upon an electromagnet.

Circuit breaker

Circuit breakers also use electromagnets. The circuit breaker in Figure 20.16 uses electromagnetism to cut off the current if it becomes larger than a certain value. If the current is too high the electromagnet becomes strong enough to pull the iron catch out of position so that the contacts open and the circuit breaks. Once the problem in the circuit has been corrected the catch is repositioned by pressing the reset button.



▲ Figure 20.16 a The circuit breaker allows a normal sized current. b When there is too large a current the circuit becomes incomplete.



▲ Figure 20.17 These circuit breakers control the supply of electricity to different circuits in a building. The circuit breaker will 'trip' if there is something wrong with the circuit and the current is too high.

CHAPTER QUESTIONS

More questions on electromagnetism can be found at the end of Unit 6 on page 217.

SKILLS CRITICAL THINKING (4)

1 Which of these objects are made from magnetic materials?

- a copper pipe
- b plastic paper clip
- c steel nail
- d glass bottle
- e iron bar
- f rubber tyre
- g nickel coins

SKILLS INTERPRETATION, REASONING

2 Draw a diagram of two bar magnets placed so that two of their poles are a repelling, and b attracting. Explain why you have drawn the magnets in these positions.

SKILLS CRITICAL THINKING (3)

3 a What three features of a magnetic field do field lines show?

b What is a uniform magnetic field?

SKILLS INTERPRETATION (4)

c Draw a diagram of a uniform field between two bar magnets.

SKILLS INTERPRETATION

SKILLS REASONING

7

4 a Draw a diagram to show the shape of the magnetic field around a solenoid.

b How does your diagram show where the magnetic field is strong or weak?

c Explain what happens to the magnetic field if the direction of the current through the solenoid is reversed.

5 In 1819 a scientist named Hans Christian Oersted was using a cell to produce a current in a wire. Close to the wire there was a compass. When there was a current in the wire, Oersted noticed – much to his surprise – that the compass needle moved.

a Why did the compass needle move?

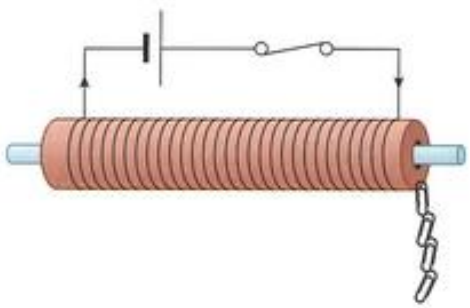
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b When there is a current in a horizontal wire, the needle of a compass placed beneath the wire comes to rest at right angles to the wire, pointing from left to right. In which direction will the compass needle point if it is held above the wire? Explain your answer.

c Would your answer to part b still be correct if the direction of the current in the wire was changed? Explain your answer.

7

6 The diagram below shows a simple electromagnet.



a Explain why the core (centre) of an electromagnet is made of soft iron and not steel.

b Suggest two ways in which the strength of this electromagnet could be increased.